Q12: The Boundaries of an LLM's Knowledge

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October 2026

Two memories

- **Parametric:** facts stored in weights; cheap, lossy, frozen at training time
- **In-context:** text in the prompt (incl. retrieved docs); exact, costs tokens
- Closed-book QA: a surprising amount *is* in the weights [**?**] — but reliability is frequency-dependent

Frequency is the master variable

- Long-tail effect: frequent facts recalled, tail facts mostly not [**?**]
- Entity popularity predicts closed-book accuracy [**?**]
- The hallucination regime is the tail: fluent distribution, no reliable memory (Q11)

The cutoff: facts vs. skills

- Post-cutoff facts: absent, full stop
- Post-cutoff skills: mostly fine — procedures generalize [**?**]
- Ask "what changed since the cutoff?" of facts, not abilities

Knowing the boundary

- Probes: better-calibrated than expected [**?**] — but calibration $\neq$ abstention
- Models answer unanswerable questions; abstention needs training [**?**]
- Probing pitfall: present $\neq$ used (Q13)

The dark twin: memorization

- Very frequent / near-duplicate text can be regurgitated verbatim [**?**]
- Non-monotone boundary: tail fabricates, extreme head leaks
- Curation is knowledge management (Q09)

Fine-tuning doesn't add knowledge cleanly

- Fine-tuning on new knowledge $\rightarrow$ more hallucination on held-out questions [**?**]
- Rule: context for *knowing*, tuning for *behaving* (Q26)

The boundary map

- Head: answer directly (parametric recall reliable)
- Tail: retrieve — flips the accuracy curve flat [**?**]
- Adaptive retrieval (only for rare entities) captures most of the benefit
- Post-cutoff: context or tools only
- Unanswerable: expect answers unless trained to abstain

Summary

A model knows what its data repeated, forgets what it omitted, cannot know what postdates its cutoff (though skills persist), and does not volunteer the boundary — so measure it and move tail questions into context.